

Code for microbiome section report:

#the function of this code is to activate the program we are using

```
conda activate qiime2
```

#Since we were already given a demux file we were able to go straight into trimming our quality plot. This code tells the program what quality reads to cut off and what quality reads to keep.

```
qiime dada2 denoise-paired \  
--i-demultiplexed-seqs demux.qza \  
--p-trunc-len-f 220 \  
--p-trim-left-f 5 \  
--p-trunc-len-r 160 \  
--p-trim-left-r 15 \  
--o-representative-sequences rep-seqs.qza \  
--o-table table.qza \  
--o-denoising-stats stats.qza
```

#This code is asking qiime2 to make a visualization file that summarizes the feature table and it creates a table of all representative sequences

```
# qiime feature-table summarize \  
--i-table table.qza \  
--o-visualization table.qzv \  
--m-sample-metadata-file metadata.txt  
qiime feature-table tabulate-seqs \  
--i-data rep-seqs.qza \  
--o-visualization rep-seqs.qzv
```

#This code is asking to download the taxonomy data to the 16s sequences that we provided.

```
wget \  

```

```
-O "gg-13-8-99-515-806-nb-classifier.qza" \
```

```
"https://data.qiime2.org/classifiers/sklearn-1.4.2/greengenes/gg-13-8-99-515-806-nb-classifier.qza"
```

#this code is asking it to assign the corresponding taxonomy to their respective 16s sequences

```
qiime feature-classifier classify-sklearn \  
--i-classifier gg-13-8-99-515-806-nb-classifier.qza \  
--i-reads rep-seqs.qza \  
--o-classification taxonomy.qza
```

#this code is asking to create a visual table of the assigned taxonomy it just created

```
qiime metadata tabulate \  
--m-input-file taxonomy.qza \  
--o-visualization taxonomy.qzv
```

#this code is asking the program to filter the feature table based on taxonomy and remove potential contaminants like mitochondria and chloroplasts

```
qiime taxa filter-table \  
--i-table table.qza \  
--i-taxonomy taxonomy.qza \  
--p-include p__ \  
--p-exclude mitochondria,chloroplast \  
--o-filtered-table  
table-with-phyla-no-mitochondria-chloroplast.qza
```

#this code is asking it to do the same as the first one but this time it's asking it to remove archaea

```
qiime taxa filter-table \  
--i-table table-with-phyla-no-mitochondria-chloroplast.qza \  
--i-taxonomy taxonomy.qza \
```

```
--p-exclude "k__Archaea" \
```

```
--o-filtered-table
```

```
table-with-phyla-no-mitochondria-chloroplasts-archaea.qza
```

#this code is also filtering the table but asking it to remove eukaryotic features this time.

```
qiime taxa filter-table \
```

```
--i-table
```

```
table-with-phyla-no-mitochondria-chloroplasts-archaea.qza \
```

```
--i-taxonomy taxonomy.qza \
```

```
--p-exclude "k__Eukaryota" \
```

```
--o-filtered-table
```

```
table-with-phyla-no-mitochondria-chloroplasts-archaea-eukaryota.  
qza
```

#this code is also filtering but this time it is filtering the representative sequences keeping only the ones with known phyla and removing contaminants like mitochondria and chloroplasts.

```
qiime taxa filter-seqs \
```

```
--i-sequences rep-seqs.qza \
```

```
--i-taxonomy taxonomy.qza \
```

```
--p-include p__ \
```

```
--p-exclude mitochondria,chloroplast \
```

```
--o-filtered-sequences
```

```
rep-seqs-with-phyla-no-mitochondria-chloroplast.qza
```

#this code is filtering the representative sequences but excluding the archaea

```
qiime taxa filter-seqs \
```

```
--i-sequences
```

```
rep-seqs-with-phyla-no-mitochondria-chloroplast.qza \
```

```
--i-taxonomy taxonomy.qza \
```

```
--p-exclude "k__Archaea" \
```

```
--o-filtered-sequences  
rep-seqs-with-phyla-no-mitochondria-chloroplasts-archaea.qza
```

#this code is also filtering the representative sequences but now its excluding any eukaryotic features

```
qiime taxa filter-seqs \  
--i-sequences  
rep-seqs-with-phyla-no-mitochondria-chloroplasts-archaea.qza \  
--i-taxonomy taxonomy.qza \  
--p-exclude "k__Eukaryota" \  
--o-filtered-sequences  
rep-seqs-with-phyla-no-mitochondria-chloroplasts-archaea-eukaryo  
ta.qza
```

#this code is renaming the files that were just cleaned up into “filtered-table.qza” and “filtered-rep-seqs.qza

```
mv  
table-with-phyla-no-mitochondria-chloroplasts-archaea-eukaryota.  
qza filtered-table.qza
```

```
mv  
rep-seqs-with-phyla-no-mitochondria-chloroplasts-archaea-eukaryo  
ta.qza filtered-rep-seqs.qza
```

#this code is asking the program to create a barplot out of the filtered table that we just created

```
qiime taxa barplot \  
--i-table filtered-table.qza \  
--i-taxonomy taxonomy.qza \  
--m-metadata-file metadata.txt \  
--o-visualization taxa-bar-plots.qzv
```

#this code is asking the program to make a phylogeny out of the filtered-rep-seqs file and it creates both a rooted and unrooted tree

```
qiime phylogeny align-to-tree-mafft-fasttree \  

```

```
--i-sequences filtered-rep-seqs.qza \  
--o-alignment aligned-rep-seqs.qza \  
--o-masked-alignment masked-aligned-rep-seqs.qza \  
--o-tree unrooted-tree.qza \  
--o-rooted-tree rooted-tree.qza
```

#this code makes a summary of the filtered table that we created earlier as well as a visualization for it

```
qiime feature-table summarize \  
--i-table filtered-table.qza \  
--o-visualization filtered-table.qzv \  
--m-sample-metadata-file metadata.txt
```

#this code is asking the program to run both alpha and beta diversity metrics but will only keep the lowest sampling number that I chose, in this case I chose 8,000

```
qiime diversity core-metrics-phylogenetic --i-phylogeny  
rooted-tree.qza --i-table filtered-table.qza --p-sampling-depth  
8000 --m-metadata-file metadata.txt --output-dir  
diversity-metrics-results
```

this code is asking the program to run alpha diversity metrics programs. This specific code is asking it to run observed features which will compare how many different microbes there are within the sample

```
qiime diversity alpha-group-significance \  
--i-alpha-diversity  
diversity-metrics-results/observed_features_vector.qza \  
--m-metadata-file metadata.txt \  
--o-visualization  
diversity-metrics-results/observed-features-group-significance.q  
zv
```

#this code is asking it to do the same alpha diversity metrics but this time using shannon

```
qiime diversity alpha-group-significance \  
--i-alpha-diversity diversity-metrics-results/shannon_vector.qza \  
--m-metadata-file metadata.txt \  
--o-visualization \  
diversity-metrics-results/shannon-group-significance.qzv
```

#this code is asking the program to test if there is any statistically different samples between groups in the population metadata column using the bray-curtis program

```
qiime diversity beta-group-significance \  
--i-distance-matrix \  
diversity-metrics-results/bray_curtis_distance_matrix.qza \  
--m-metadata-file metadata.txt \  
--m-metadata-column population \  
--o-visualization \  
diversity-metrics-results/bray-curtis-population-significance.qzv \  
--p-pairwise
```

#this code is asking the bray curtis program to look for any statistically different data in the sex metadata column.

```
qiime diversity beta-group-significance \  
--i-distance-matrix \  
diversity-metrics-results/bray_curtis_distance_matrix.qza \  
--m-metadata-file metadata.txt \  
--m-metadata-column sex \  
--o-visualization \  
diversity-metrics-results/bray-curtis-sex-significance.qzv \  

```

```
--p-pairwise
```

#this code is asking to find any statistical difference between the composition of the microbes in the population using a different program called Weighted Unifrac

```
qiime diversity beta-group-significance \  
  
--i-distance-matrix  
diversity-metrics-results/weighted_unifrac_distance_matrix.qza \  
  
--m-metadata-file metadata.txt \  
  
--m-metadata-column population \  
  
--o-visualization  
diversity-metrics-results/weighted-unifrac-population-significan  
ce.qzv \  
  
--p-pairwise
```

#this code is asking to find any statistical difference between the composition of the microbes in gender (sex) using a different program called Weighted Unifrac

```
qiime diversity beta-group-significance \  
  
--i-distance-matrix  
diversity-metrics-results/weighted_unifrac_distance_matrix.qza \  
  
--m-metadata-file metadata.txt \  
  
--m-metadata-column sex \  
  
--o-visualization  
diversity-metrics-results/weighted-unifrac-sex-significance.qzv \  
  
--p-pairwise
```

#this code is asking the program to run ANCOM-BC to identify differentially abundant microbial features in the sex. I chose to only do sex because that's what showed statistical differences in the previous beta diversity tests.

```
qiime composition ancombc \  
  
--i-table filtered-table.qza \  
  
--m-metadata-file metadata.txt \  
  
--o-visualization  
composition-ancombc-sex-significance.qzv
```

```
--p-formula 'sex' \
```

```
--o-differentials ancombc-sex.qza
```

#this code is asking the program to make an abundance barplot to visually display what we asked it to find in the previous code.

```
qiime composition da-barplot \
```

```
--i-data ancombc-sex.qza \
```

```
--p-significance-threshold 0.05 \
```

```
--o-visualization da-barplot-sex.qzv
```